1.Data Import

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Configure global plot aesthetics for better readability
sns.set_style('whitegrid')
plt.rcParams['figure.dpi'] = 110

# Load the crime dataset into a pandas DataFrame
df = pd.read_csv('crime.csv')

# Output the total number of rows and columns to verify the data size
print("Dataset shape:", df.shape)

# Display the first few rows of the dataset
print(df.head())
```

```
   B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Total Offences under IPC r/w IT Act
0                                                  0
1                                                 42
2                                                 30
3                                                  0
4                                                 14

   C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Gambling Act (Online Gambling) - Col
0                                                  0
1                                                  0
2                                                  0
3                                                  0
4                                                  0

   C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Lotteries Act (Online Lotteries) - C
0                                                  0
1                                                  0
2                                                  0
3                                                  0
4                                                  0

   C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Copy Right Act, 1957' - Col. ( 47)  '
0                                                  0
1                                                  0
2                                                  0
3                                                  0
4                                                  0

   C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Trade Marks Act, 1999 - Col. ( 48)  '
0                                                  0
1                                                  0
2                                                  0
3                                                  0
4                                                  0

   C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Other SLL Crimes - Col. ( 49)  \
0                                                  0
1                                                  0
2                                                  0
3                                                  0
4                                                  0

   C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Total Offences under SLL r/w IT Act
0                                                  0
1                                                  0
2                                                  0
3                                                  0
4                                                  0

   Total Cyber Crimes (A+B+C) - Col. ( 51)
0                                       1
1                                      55
2                                      38
3                                       5
4                                      17

[5 rows x 52 columns]
```

2. Initial Inspection

```
# Display a concise summary of the DataFrame, including column names, non-null counts, and data types
df.info()

# Print a blank line to separate the outputs for better readability
print()

# Calculate and display the total number of missing (NaN) values present in each column
print("Missing values per column:")
print(df.isnull().sum())
print()

# Count and display the total number of exact duplicate rows in the dataset
print("Number of duplicate rows:", df.duplicated().sum())
```

```
Missing values per column:
Sl. No
State/UT
District
A. Offences under I.T.Act - Tampering computer source documents (Sec.65) - Col. ( 3)
A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Total) (Col.5+Col.8+Col.9+Col.10+Col.11) - (
A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Computer Related Offences (Sec.66
A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Ransom-ware - Col. ( 6)
A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Offences other than Ransom-ware -
A. Offences under I.T.Act - Computer Related Offences - Dishonestly receiving stolen computer resource or communication device
A. Offences under I.T.Act - Computer Related Offences - Identity Theft (Sec.66C) - Col. ( 9)
A. Offences under I.T.Act - Computer Related Offences - Cheating by personation by using computer resource (Sec.66D) - Col. ( 1
A. Offences under I.T.Act - Computer Related Offences - Violation of Privacy (Sec.66E) - Col. ( 11)
A. Offences under I.T.Act - Cyber Terrorism (Sec.66 F) - Col. ( 12)
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Publicat
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Publish
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Publish
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Publish
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Preserva
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Other Se
A. Offences under I.T.Act - Inter-ception or Monitoring or decryption of Information (Sec.69) - Col. ( 19)
A. Offences under I.T.Act - Un-authorized access/attempt to access to protected computer system (Sec.70) - Col. ( 20)
A. Offences under I.T.Act - Abetment to Commit Offences (Sec.84 B) - Col. ( 21)
A. Offences under I.T.Act - Attempt to Commit Offences (Sec.84C) - Col. ( 22)
A. Offences under I.T.Act - Other Sections of IT Act - Col. ( 23)
A. Offences under I.T.Act - Total Offences under I.T. Act (Col.3+Col.4+Col.12+Col.13+Col.19+Col.20+Col.21+Col.22+Col.23) - Col
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Abetment of Suicide (Online) (Sec.305/3(
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Cyber Stalking/ Bullying of Women/ Chil(
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Data theft (Sec.379 to 381) - Col. ( 27
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Cheating (Sec.420) - Col. ( 34)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Forgery (Sec.465, 468 & 471) - Col. ( 3
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Defamation/ Morphing (Sec.469 IPC r/w IT
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fake Profile (r/w IPC/SLL) - Col. ( 37)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Counterfeiting - Counter-feiting (Total
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Counterfeiting - Currency (Sec.489A to 4
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Counterfeiting - Stamps (Sec.255) - Col
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Cyber Blackmailing/Threatening (Sec.506
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fake News on Social Media (Sec. 505) - (
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Other Offences (r/w IT Act) - Col. ( 43
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Total Offences under IPC r/w IT Act (Co
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Gambling Act (Online Gambling) - Col. (
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Lotteries Act (Online Lotteries) - Col.
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Copy Right Act, 1957' - Col. ( 47)
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Trade Marks Act, 1999 - Col. ( 48)
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Other SLL Crimes - Col. ( 49)
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Total Offences under SLL r/w IT Act (Tot
Total Cyber Crimes (A+B+C) - Col. ( 51)
dtype: int64

Number of duplicate rows: 0
```

3. Data Cleaning

3.1 Handling Missing Values

```
# Impute missing values in continuous numeric columns using their respective medians
columns_to_impute = [
    'A. Offences under I.T.Act - Cyber Terrorism (Sec.66 F) - Col. ( 12)',
    'C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Gambling Act (Online Gambling) - Cc
    'Total Cyber Crimes (A+B+C) - Col. ( 51)'
]

for col in columns_to_impute:
    median_val = df[col].median()
    # Count missing values before imputation to log what is being changed
    n_missing = df[col].isnull().sum()
    # Impute (fill) any missing values in the column with its calculated median
    df[col] = df[col].fillna(median_val)
    print(f"{col[:30]}... : filled {n_missing} missing values with median = {median_val:.2f}")

# Check if any missing values remain elsewhere in the broader dataset
print("\nMissing values remaining:", df.isnull().sum().sum())
```

```
A. Offences under I.T.Act - Cy... : filled 0 missing values with median = 0.00
C. Offences under SLL (Involvi... : filled 0 missing values with median = 0.00
Total Cyber Crimes (A+B+C) - C... : filled 0 missing values with median = 17.00

Missing values remaining: 0
```

3.2 Removing Duplicate Rows

```
# Remove any exact duplicate rows and log the total number dropped
before = len(df)
df = df.drop_duplicates().reset_index(drop=True)

# Record the total number of rows after the drop operation
after = len(df)
print(f"Removed {before - after} duplicate rows.")

# Print the final shape of the dataset to confirm its current dimensions.
print(f"Dataset shape after removing duplicates: {df.shape}")
```

```
Removed 0 duplicate rows.
Dataset shape after removing duplicates: (970, 52)
```

3.3 Handling Outliers

```
# Define a helper function to calculate Interquartile Range (IQR) boundaries
def iqr_bounds(series):
    # Calculate the 25th percentile (Q1) and 75th percentile (Q3)
    Q1 = series.quantile(0.25)
    Q3 = series.quantile(0.75)
    IQR = Q3 - Q1
    return Q1 - 1.5 * IQR, Q3 + 1.5 * IQR

# Select continuous numeric columns from the crime dataset to check for outliers
crime_cols_to_check = [
    'A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Total) (Col.5+Col.8+Col.9+Col.10+Col.11
    'Total Cyber Crimes (A+B+C) - Col. ( 51)'
]

# Detect extreme values in numerical columns and cap them at IQR boundaries (winsorization)
for col in crime_cols_to_check:
    # Unpack the calculated lower and upper bounds for the current column
    low, high = iqr_bounds(df[col])
    n_outliers = ((df[col] < low) | (df[col] > high)).sum()

    # Using string slicing col[:40] just to keep the print statement readable in the console
    print(f"{col[:40]}...: IQR bounds = [{low:.2f}, {high:.2f}], outliers found = {n_outliers}")

    # Apply winsorization: cap any extreme values at the exact lower and upper boundaries.
    df[col] = df[col].clip(lower=low, upper=high)
```

```
print("\nOutliers capped to IQR bounds (winsorized).")

# Display summary statistics to verify the final data distribution.
# Notice that the 'max' values will now perfectly align with the calculated upper bounds.
print(df[crime_cols_to_check].describe())
```

```
A. Offences under I.T.Act - Computer Rel...: IQR bounds = [-16.50, 27.50], outliers found = 141
Total Cyber Crimes (A+B+C) - Col. ( 51)...: IQR bounds = [-73.50, 130.50], outliers found = 77

Outliers capped to IQR bounds (winsorized).
       A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Total) (Col.5+Col.8+Col.9+Col.10+Col.1
count                                          970.000000
mean                                             6.845876
std                                             10.142036
min                                              0.000000
25%                                              0.000000
50%                                              1.000000
75%                                             11.000000
max                                             27.500000

       Total Cyber Crimes (A+B+C) - Col. ( 51)
count                                970.000000
mean                                  34.666495
std                                   40.156510
min                                    0.000000
25%                                    3.000000
50%                                   17.000000
75%                                   54.000000
max                                  130.500000
```

4. Exploratory Data Analysis (EDA)

4.1 Descriptive Statistics

```
# View basic statistical details like mean, min, and max for numerical columns
print(df.describe())
```

```
       A. Offences under I.T.Act - Tampering computer source documents (Sec.65) - Col. ( 3)  \
count                                          970.000000
mean                                             0.134021
std                                              1.309084
min                                              0.000000
25%                                              0.000000
50%                                              0.000000
75%                                              0.000000
max                                             26.000000

       A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Total) (Col.5+Col.8+Col.9+Col.10+Col
count                                          970.000000
mean                                             6.845876
std                                             10.142036
min                                              0.000000
25%                                              0.000000
50%                                              1.000000
75%                                             11.000000
max                                             27.500000

       A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Computer Related Offences
count                                          970.000000
mean                                             7.973196
std                                            104.027122
min                                              0.000000
25%                                              0.000000
50%                                              0.000000
75%                                              0.000000
max                                           3031.000000

       A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Ransom-ware - Col. ( 6)  \
count                                          970.000000
mean                                             1.975258
std                                             27.313216
min                                              0.000000
25%                                              0.000000
50%                                              0.000000
75%                                              0.000000
max                                            681.000000

       A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Offences other than Ransom
```

```
count                                        970.000000
mean                                           5.997938
std                                           81.964905
min                                            0.000000
25%                                            0.000000
50%                                            0.000000
75%                                            0.000000
max                                         2350.000000

       A. Offences under I.T.Act - Computer Related Offences - Dishonestly receiving stolen computer resource or communication
count                                        970.000000
mean                                           0.738144
std                                            7.010643
min                                            0.000000
25%                                            0.000000
50%                                            0.000000
```

4.2 Correlation Analysis

```
# Calculate and display the correlation matrix for specific numerical columns
# Using primary crime categories from the dataset
numeric_cols = [
    'A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Total) (Col.5+Col.8+Col.9+Col.10+Col.11
    'B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471
    'Total Cyber Crimes (A+B+C) - Col. ( 51)'
]

corr_matrix = df[numeric_cols].corr()
print(corr_matrix.round(5))
```

```
                                                   A. Offences under I.T.Act - Computer Related Offences - Computer Related Off
A. Offences under I.T.Act - Computer Related Of...                                                                        1.00000
B. Offences under IPC (Involving Communication ...                                                                        0.12782
Total Cyber Crimes (A+B+C) - Col. ( 51)                                                                                   0.63043

                                                   B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w
A. Offences under I.T.Act - Computer Related Of...                                                                        0.12782
B. Offences under IPC (Involving Communication ...                                                                        1.00000
Total Cyber Crimes (A+B+C) - Col. ( 51)                                                                                   0.21727

                                                   Total Cyber Crimes (A+B+C) - Col. ( 51)
A. Offences under I.T.Act - Computer Related Of...                                 0.63043
B. Offences under IPC (Involving Communication ...                                 0.21727
Total Cyber Crimes (A+B+C) - Col. ( 51)                                            1.00000
```

4.3 Average Cyber Crimes and IT Offences by State

```
# Group by State to calculate average crime metrics, sorting by highest average total crimes
state_summary = df.groupby('State/UT').agg(
    Avg_Total_Crimes=('Total Cyber Crimes (A+B+C) - Col. ( 51)', 'mean'),
    Avg_IT_Offences=('A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Total) (Col.5+Col.8+Cc
    District_Count=('District', 'count')
).round(1).sort_values('Avg_Total_Crimes', ascending=False)

# Display the top 10 states
print(state_summary.head(10))
```

```
                Avg_Total_Crimes  Avg_IT_Offences  District_Count
State/UT
Total Districts              89.0             19.9              36
Telangana                    85.5              2.0              30
Uttar Pradesh                73.4             20.3              78
Maharashtra                  58.2              6.2              49
Karnataka                    56.6             21.2              38
Andhra Pradesh               55.7              7.3              28
Odisha                       52.1              4.1              37
Tamil Nadu                   39.6             14.9              49
Rajasthan                    38.1              5.2              48
Jharkhand                    35.4             16.5              26
```

4.4 Interactive Data Visualization

```
import pandas as pd
```

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

plt.rcParams['figure.figsize']=(10,6)

# Load the dataset
df = pd.read_csv('crime.csv')

# Rename the extremely long NCRB columns to simple names for clean visualizations
df = df.rename(columns={
    'A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Total) (Col.5+Col.8+Col.9+Col.10+Col.11
    'B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471
    'Total Cyber Crimes (A+B+C) - Col. ( 51)': 'Total_Cyber_Crimes'
})

df.head()
print("Dataset Shape:", df.shape)
df.info()
print(df.dtypes)
df.describe()
print(df.isnull().sum())
print("Duplicates:", df.duplicated().sum())

# Drop duplicates if any
df = df.drop_duplicates()

# 1. Categorical Countplots (Using State/UT)
categorical_cols = ['State/UT']

for col in categorical_cols:
    plt.figure(figsize=(10,6))
    sns.countplot(data=df, x=col, palette='viridis')
    plt.title(f'Distribution of Districts by {col}')
    plt.xticks(rotation=90) # Rotated 90 degrees so the 36 state names don't overlap
    plt.tight_layout()
    plt.show()

# 2. Numerical Histograms
numerical_cols = ['Total_Cyber_Crimes', 'IT_Act_Offences', 'Online_Fraud']
df[numerical_cols].hist(bins=30, figsize=(15,5), color='skyblue')
plt.tight_layout()
plt.show()

# 3. Scatter Plot (IT Offences vs Total Crimes, color-coded by State)
sns.scatterplot(data=df, x='IT_Act_Offences', y='Total_Cyber_Crimes', hue='State/UT', legend=False)
plt.title('Total Cyber Crimes vs IT Act Offences')
plt.show()

# 4. Box Plot (Total Crimes Distribution by State)
sns.boxplot(data=df, x='State/UT', y='Total_Cyber_Crimes', palette='winter')
plt.title('Total Cyber Crimes Distribution by State/UT')
plt.xticks(rotation=90) # Rotate state names
plt.tight_layout()
plt.show()

# 5. Correlation Heatmap
corr_matrix = df[numerical_cols].corr(numeric_only=True)
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap of Cyber Crimes')
plt.show()

# 6. Interactive Plotly Scatter (Hover shows the exact District)
fig = px.scatter(df, x='IT_Act_Offences', y='Total_Cyber_Crimes', color='State/UT',
                 hover_data=['District', 'Online_Fraud'],
                 title='Interactive: Total Cyber Crimes vs IT Act Offences')
fig.show()
```

```
Dataset Shape: (970, 52)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 970 entries, 0 to 969
Data columns (total 52 columns):
 #   Column
---  ------
 0   Sl. No
 1   State/UT
 2   District
 3   A. Offences under I.T.Act - Tampering computer source documents (Sec.65) - Col. ( 3)
 4   IT_Act_Offences
 5   A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Computer Related Offences (Se
 6   A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Ransom-ware - Col. ( 6)
 7   A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Offences other than Ransom-wa
 8   A. Offences under I.T.Act - Computer Related Offences - Dishonestly receiving stolen computer resource or communication de
 9   A. Offences under I.T.Act - Computer Related Offences - Identity Theft (Sec.66C) - Col. ( 9)
 10  A. Offences under I.T.Act - Computer Related Offences - Cheating by personation by using computer resource (Sec.66D) - Co
 11  A. Offences under I.T.Act - Computer Related Offences - Violation of Privacy (Sec.66E) - Col. ( 11)
 12  A. Offences under I.T.Act - Cyber Terrorism (Sec.66 F) - Col. ( 12)
 13  A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Pub
 14  A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Pub
 15  A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Pub
 16  A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Pub
 17  A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Pre
 18  A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Oth
 19  A. Offences under I.T.Act - Inter-ception or Monitoring or decryption of Information (Sec.69) - Col. ( 19)
 20  A. Offences under I.T.Act - Un-authorized access/attempt to access to protected computer system (Sec.70) - Col. ( 20)
 21  A. Offences under I.T.Act - Abetment to Commit Offences (Sec.84 B) - Col. ( 21)
 22  A. Offences under I.T.Act - Attempt to Commit Offences (Sec.84C) - Col. ( 22)
 23  A. Offences under I.T.Act - Other Sections of IT Act - Col. ( 23)
 24  A. Offences under I.T.Act - Total Offences under I.T. Act (Col.3+Col.4+Col.12+Col.13+Col.19+Col.20+Col.21+Col.22+Col.23)
 25  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Abetment of Suicide (Online) (Sec.3
 26  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Cyber Stalking/ Bullying of Women/
 27  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Data theft (Sec.379 to 381) - Col.
 28  Online_Fraud
 29  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471
 30  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471
 31  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471
 32  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471
 33  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471
 34  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Cheating (Sec.420) - Col. ( 34)
 35  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Forgery (Sec.465, 468 & 471) - Col
 36  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Defamation/ Morphing (Sec.469 IPC
 37  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fake Profile (r/w IPC/SLL) - Col.
 38  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Counterfeiting - Counter-feiting (
 39  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Counterfeiting - Currency (Sec.489A
 40  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Counterfeiting - Stamps (Sec.255)
 41  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Cyber Blackmailing/Threatening (Sec
 42  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fake News on Social Media (Sec. 50
 43  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Other Offences (r/w IT Act) - Col.
 44  B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Total Offences under IPC r/w IT Act
 45  C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Gambling Act (Online Gambling) - Co
 46  C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Lotteries Act (Online Lotteries) -
 47  C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Copy Right Act, 1957' - Col. ( 47)
 48  C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Trade Marks Act, 1999 - Col. ( 48)
 49  C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Other SLL Crimes - Col. ( 49)
 50  C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Total Offences under SLL r/w IT Act
 51  Total_Cyber_Crimes
dtypes: int64(49), object(3)
memory usage: 394.2+ KB
Sl. No
State/UT
District
A. Offences under I.T.Act - Tampering computer source documents (Sec.65) - Col. ( 3)
IT_Act_Offences
A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Computer Related Offences (Sec.66
A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Ransom-ware - Col. ( 6)
A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Offences other than Ransom-ware -
A. Offences under I.T.Act - Computer Related Offences - Dishonestly receiving stolen computer resource or communication device
A. Offences under I.T.Act - Computer Related Offences - Identity Theft (Sec.66C) - Col. ( 9)
A. Offences under I.T.Act - Computer Related Offences - Cheating by personation by using computer resource (Sec.66D) - Col. ( 1
A. Offences under I.T.Act - Computer Related Offences - Violation of Privacy (Sec.66E) - Col. ( 11)
A. Offences under I.T.Act - Cyber Terrorism (Sec.66 F) - Col. ( 12)
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Publicat
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Publish
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Publish
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Publish
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Preserva
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Other Se
A. Offences under I.T.Act - Inter-ception or Monitoring or decryption of Information (Sec.69) - Col. ( 19)
A. Offences under I.T.Act - Un-authorized access/attempt to access to protected computer system (Sec.70) - Col. ( 20)
A. Offences under I.T.Act - Abetment to Commit Offences (Sec.84 B) - Col. ( 21)
A. Offences under I.T.Act - Attempt to Commit Offences (Sec.84C) - Col. ( 22)
A. Offences under I.T.Act - Other Sections of IT Act - Col. ( 23)
```

```
A. Offences under I.T.Act - Total Offences under I.T. Act (Col.3+Col.4+Col.12+Col.13+Col.19+Col.20+Col.21+Col.22+Col.23) - Col
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Abetment of Suicide (Online) (Sec.305/3(
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Cyber Stalking/ Bullying of Women/ Child
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Data theft (Sec.379 to 381) - Col. ( 27
Online_Fraud
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Cheating (Sec.420) - Col. ( 34)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Forgery (Sec.465, 468 & 471) - Col. ( 3
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Defamation/ Morphing (Sec.469 IPC r/w IT
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fake Profile (r/w IPC/SLL) - Col. ( 37)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Counterfeiting - Counter-feiting (Total
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Counterfeiting - Currency (Sec.489A to 4
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Counterfeiting - Stamps (Sec.255) - Col
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Cyber Blackmailing/Threatening (Sec.506
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fake News on Social Media (Sec. 505) - (
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Other Offences (r/w IT Act) - Col. ( 43
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Total Offences under IPC r/w IT Act (Co
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Gambling Act (Online Gambling) - Col. (
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Lotteries Act (Online Lotteries) - Col.
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Copy Right Act, 1957' - Col. ( 47)
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Trade Marks Act, 1999 - Col. ( 48)
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Other SLL Crimes - Col. ( 49)
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Total Offences under SLL r/w IT Act (Tot
Total_Cyber_Crimes
dtype: object
Sl. No
State/UT
District
A. Offences under I.T.Act - Tampering computer source documents (Sec.65) - Col. ( 3)
IT_Act_Offences
A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Computer Related Offences (Sec.66
A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Ransom-ware - Col. ( 6)
A. Offences under I.T.Act - Computer Related Offences - Computer Related Offences (Sec.66) - Offences other than Ransom-ware -
A. Offences under I.T.Act - Computer Related Offences - Dishonestly receiving stolen computer resource or communication device
A. Offences under I.T.Act - Computer Related Offences - Identity Theft (Sec.66C) - Col. ( 9)
A. Offences under I.T.Act - Computer Related Offences - Cheating by personation by using computer resource (Sec.66D) - Col. ( 1
A. Offences under I.T.Act - Computer Related Offences - Violation of Privacy (Sec.66E) - Col. ( 11)
A. Offences under I.T.Act - Cyber Terrorism (Sec.66 F) - Col. ( 12)
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Publicat
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Publish
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Publish
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Publish
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Preserva
A. Offences under I.T.Act - Publication/transmission of obscene / sexually explicit act in electronic form (Sec. 67) - Other Se
A. Offences under I.T.Act - Inter-ception or Monitoring or decryption of Information (Sec.69) - Col. ( 19)
A. Offences under I.T.Act - Un-authorized access/attempt to access to protected computer system (Sec.70) - Col. ( 20)
A. Offences under I.T.Act - Abetment to Commit Offences (Sec.84 B) - Col. ( 21)
A. Offences under I.T.Act - Attempt to Commit Offences (Sec.84C) - Col. ( 22)
A. Offences under I.T.Act - Other Sections of IT Act - Col. ( 23)
A. Offences under I.T.Act - Total Offences under I.T. Act (Col.3+Col.4+Col.12+Col.13+Col.19+Col.20+Col.21+Col.22+Col.23) - Col
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Abetment of Suicide (Online) (Sec.305/3(
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Cyber Stalking/ Bullying of Women/ Child
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Data theft (Sec.379 to 381) - Col. ( 27
Online_Fraud
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fraud (Sec.420 r/w Sec.465,468-471 IPC)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Cheating (Sec.420) - Col. ( 34)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Forgery (Sec.465, 468 & 471) - Col. ( 3
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Defamation/ Morphing (Sec.469 IPC r/w IT
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fake Profile (r/w IPC/SLL) - Col. ( 37)
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Counterfeiting - Counter-feiting (Total
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Counterfeiting - Currency (Sec.489A to 4
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Counterfeiting - Stamps (Sec.255) - Col
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Cyber Blackmailing/Threatening (Sec.506
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Fake News on Social Media (Sec. 505) - (
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Other Offences (r/w IT Act) - Col. ( 43
B. Offences under IPC (Involving Communication Devices as Medium/Target) r/w IT Act. - Total Offences under IPC r/w IT Act (Co
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Gambling Act (Online Gambling) - Col. (
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Lotteries Act (Online Lotteries) - Col.
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Copy Right Act, 1957' - Col. ( 47)
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Trade Marks Act, 1999 - Col. ( 48)
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Other SLL Crimes - Col. ( 49)
C. Offences under SLL (Involving Communication Devices as Medium/Target) r/w IT Act. - Total Offences under SLL r/w IT Act (Tot
Total_Cyber_Crimes
dtype: int64
Duplicates: 0
/tmp/ipykernel_666/889235787.py:35: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and se
```





